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OCEAN RESEARCH AND DEVELOPMENT - ROLE AND FUNCTIONS

Introduction

India is endowed with a well-established maritime tradition. Long coastline and the adventurous nature of our ancestors took us to far off places. As far as ocean sciences are concerned, early nineteen eighties mark the beginning of a new era. The adoption of the Convention of the UN Conference on the Law of the Seas in 1982 established a new international order for the oceans. This extended the economic jurisdiction of coastal states to an area ranging from 200 to 350 miles from the coastline. According to this regime, nearly 2.02 million square kilometres of area, or nearly two-third of the land mass came under India's jurisdiction. In this area, the exclusive right to utilize living and non-living resources vests with the nation.

Department of Ocean Development (DOD):

The Department of Ocean Development (DOD) was created in July, 1981. In 1982, the Government of India issued the Ocean Policy Statement in which control, management and utilization of the internal resources available in the sea and development of appropriate technologies to harness these resources have been emphasized. From March 1982, the DOD has been functioning as a separate nodal department in line with the Ocean Policy Statement of 1982.

The programmes pursued by the Department over the years have also kept pace with developments world over and addressed national needs and issues. In addition to basic knowledge to determine the potentialities inherent in the Indian sea-space we had to develop appropriate technologies to harness these resources. A supporting infrastructure had to be built. Effective systems of management and control of the entire set-up were also necessary.

Keeping this in view, various institutional mechanisms were set up, viz. National Institute of Ocean Technology (NIOT) at Chennai, National Centre for Antarctic and

Ocean Research (NCAOR) at Goa, and Indian National Centre for Ocean Information Service at Hyderabad. Additionally, the existing institutes and science departments have significantly contributed to DOD's endeavour.

A special mention must be made of the National Institute of Oceanography (NIO), Goa, one of the CSIR institutes which has very significantly added information on several disciplines.

The Department's programmes can be broadly classified in the following groups:

1. Multi-institutional and multi-disciplinary Polar Science Programme and Antarctic Expeditions with scientific and geopolitical significance;
2. Marine Non-Living Resources Programme Minerals and Energy;
3. Marine Living Resources Programme;
4. Marine Environment and Coastal Zone;
5. Ocean Atmosphere and Climatology;
6. Capacity Building Programme Towards Self Reliance and Creation of Public Awareness of Ocean, its Potential and Usage.

The scientific research and technology development programmes mentioned above are formulated, coordinated and executed through autonomous institutions attached offices supported by the Department. These multi-disciplinary and multi-institutional programmes are also assisted by national research laboratories, academic institutions and industries.

The Department is the nodal agency for international programmes in the ocean sector and represents the country in Intergovernmental Oceanographic Commission (IOC) of UNESCO, Regional Committee of IOC in Coastal Indian Ocean (IOCINDIO), International Sea-Bed Authority (ISBA) and the State Parties of the United Nations Convention on the Law of the Seas (UNCLOS), the Antarctic Treaty System (ATS) and its scientific and managerial organs Commission on Conservation of Antarctic Marine Living Resources (CCAMLR), Council of Managers of National Antarctic Programme (COMNAP), and Scientific Committee on Antarctic Research (SCAR).

These activities have scientific, economic and geopolitical impacts. Further, marine research and capacity building is one of the key areas of the Department to promote basic research in marine science and establish centres of excellence in academic institutions.

The Mission:

To improve our understanding of the ocean, specifically the Indian Ocean, for sustainable development of ocean resources, improving livelihood, and timely warnings of coastal hazards, that will make India an exemplary steward of her people and ocean.

The Vision:

To improve our understanding of ocean processes through conceiving and implementing long-term observational programmes and incubating cutting edge marine technology so that we are able to:

i. Improve understanding of the Indian Ocean and its various inter-related processes;
ii. Assess the living and non-living resources of our seas and their sustainable levels of utilisation.
iii. Contribute to the forecast of the course of the monsoons and extreme events;
iv. Model sustainable uses of the coastal zone for decision making;
v. Forge partnerships with Indian Ocean neighbours through the awareness and concept of one ocean; and
vi. Secure recognition for the interests of India and the Indian Ocean in regional and international bodies.

Mineral Resources:

Mineral exploration will undoubtedly be influenced by recent developments in the plate tectonic theory.

Three types of environment are areas of major igneous activity:

(i) spreading plate boundaries;
(ii) subduction zones; and
(iii) hotspots of rising mantle rock.

Ore deposits have been shown to be associated with each of these zones, sea ores are often highly concentrated; and that shelves are rich in petroleum, natural gas, tin, phosphorite, diamonds, sulphur and iron. Manganese-rich nodules have been hailed as a bonanza that would help the economy of developing countries.

Items that could be mined from the sea-floor include petroleum, natural gas, coal, limestone, tin, copper, nickel, iron, and sulphur. Chief prospects for petroleum deposits are youthful margins where some basins contain thick accumulations of sediments. Oil may also occur at convergence zones and near translation faults. Asia remains the continent with greatest promise of discovery.

National Institute of Oceanography (NIO):

The National Institute of Oceanography (NIO), Goa, initiated exploration of polymetallic nodules in the Indian Ocean in 1977. The first sample was collected by the research ship Gaveshani from the Indian Ocean, in 1980. India has located few scattered deposits of polymetallic sulphides in the central Indian Ocean. Thus India joined a select club of nations—USA, UK, erstwhile USSR, Japan and Germany—who have collected nodules from the seabed.

They occur along the mid-ocean rift valleys, and those found near the Galapagos rift system contain 48 per cent sulphur, 43 per cent iron, 11 per cent copper and smaller quantities of zinc, tin, molybdenum, lead and silver.

In 1987, India became the first of the four Registered Pioneer Investors to be given exclusive rights to survey, explore and develop a mine site in the central Indian Ocean basin (CIOB) for polymetallic nodules. India is entitled to develop the resources of the deep seabed in an area of 1, 50,000 sq.km.

India as the first 'Pioneer Investor' has to undertake different developmental activities for exploring and exploitation of polymetallic nodules towards fulfilment of the obligations relating to development and transfer of technology and training in the use of technology, etc.

This programme consists of four major components, viz.,

1. Survey and exploration of nodule in the CIOB;
2. Environmental impact assessment (EIA) study at the mine site;
3. Development of technologies for mining of nodules;
4. Extraction of metals.

The Central Mechanical Engineering Research Institute (CMERI), Durgapur, has been given the responsibility of designing and developing a mining system. The technology developed for exploration and extraction of metals will also be useful for bathymetry of the EEZ and deep seabed exploration and extraction of other seabed resources such as sulphides, metalliferrous muds and phosphorites, and extraction of metals from non-conventional mineral resources such as nickel laterites through hydrometallurgical techniques.

Coastal Ocean Monitoring and Prediction System:

Coastal Ocean Monitoring and Prediction System (COMAPS) Programme are to assess the health of our seas on a long-term basis and to facilitate the pollution control authorities to plan strategies required for management of pollution related issues like prescription of appropriate disposal standards as per the assimilation capacity of the receiving water bodies, water quality criteria, etc. This is being implemented since the year 1990-91.

Two coastal research vessels, namely, Sagar Purvi and Sagar Paschimi, are used for the coastal oceanographic research programmes. These vessels are being used for pollution monitoring under COMAPS programme of DOD and also for Integrated Coastal and Marine Area Management (ICMAM) programmes. In addition to the above cruises, the vessels Sagar Paschimi and Sagar Purvi participated in the marine archaeological findings at Gulf of Cambay and multi-beam sonar survey at Tuticorin port.

The ICMAM project is being implemented from 1997-98 onwards. This project has two major components, viz., capacity building and development of infrastructure for R&D, and training.

The capacity building component, which is funded by the World Bank under an environment management capacity building project of the Ministry of Environment and Forests, has five major activities as briefly described below:

1. Development of GIS-based Information System for Critical Habitats,
2. Determination of Waste Assimilation Capacity,
3. Development of Guidelines for Environmental Impact Assessment:
4. Development of Model ICMAM Plans for Chennai, Goa and Gulf of Kachch
5. Development of Potential Drugs from the Ocean

Polar Exploration:

Antarctic Expeditions:

Several scientific expeditions have been sent to the icy continent of Antarctica by India since 1981. The objective of Antarctic research is to identify and initiate programmes of scientific and economic significance besides establishing infrastructure and expertise in the field. In December 1995, India's first commercial Antarctic expedition was launched from Kochi with the aim of catching krill for export to Japan. Sagar Sampada was used for the purpose.

From 1999 onwards the Indian scientific expeditions to Antarctica are launched from Cape Town, South Africa, resulting in mutual scientific/logistics advantages. It has also paved the way for future cooperation in the field of polar science between India and South Africa in particular and between India and other countries using Cape Town as a base for their expeditions in general.

In early 2009, an Indo-German expedition was sent Antarctica with 29 Indian researchers to conduct a two-month mega experiment to test a technique that may help fight climate change.

Dakshin Gangotri was the first station to be set up. Maitri, the second one, was established in 1988-89 on a rocky base which stays free of snow almost all year round. The position of Maitri is most suitable for the study of the 'Aurora Borealis'. The NPL, Ahmedabad, has developed an instrument to study daytime aurorae.

India is in the process of setting up another (third) station in Antarctica. The identified location of the new station is in the Larsemann Hills area or McLeod Island in its vicinity, 600 km east of the existing Maitri research station. The third base station is necessary is the second station Maitri, built in 1989, will shortly outlive its utility. The first station Dakshin Gangotri, built in 1983, was abandoned following a blizzard hit.

Dakshin Gangotri glacier, which has been monitored by Indian scientists for more than 20 years in Antarctica has been declared as an Antarctic Specially Protected Area (ASPA) by the governing body of the Antarctic Treaty in its XXVIII Antarctic Treaty Consultative Meeting (ATCM) held in Stockholm in June 2006. An Ice Core Laboratory to study the ice cores brought from the Antarctica, has become fully operational at NCAOR, Goa.

By virtue of its scientific activities in Antarctica, India was admitted as a Consultative Member of the Antarctic Treaty, 1959 in September, 1983. India became a member of the Scientific Committee on Antarctic Research in October 1984.

India also acceded to the Convention on the Conservation of Antarctic Marine Living Resources from July 17, 1985. In October 1991, the members of the Antarctic Treaty signed a significant agreement at Madrid. The agreement seeks to ban exploration of oil and other minerals in the Antarctica, known for its fragile eco-system.

There is to be protection of Antarctica's delicate flora and fauna for half-a- century. The agreement establishes procedures to assess the environmental impact of all human activity on the continent.

Scientific activities have included geological mapping of the Wohlihat massif covering an area of over 600 sq km and collection of samples for penological, geochemical, geochronological studies; study of various oceanographic parameters and of biomass distribution in different parts of the Arabian Sea, Bay of Bengal, and the Central Indian Ocean.

On July 1, 2008 India's first research station at the Arctic—Himadri—was inaugurated. Equipped with state-of-the-art facilities for year-round scientific research work, the station is located at Ny-Alesund in Norway, the northernmost permanent-human

settlement, 1,200 km from the North Pole. With Himadri, India has become the 11th country to have established a full- fledged research station at the Arctic. The others are Britain, Germany, France, Italy, China, Japan, South Korea, the Netherlands, Sweden and Norway.

Development of Potential Drugs from the Ocean:

The National Project on "Development of Potential Drugs from the Ocean" was taken up for implementation from 1990-91 with a view to harnessing the potential marine flora and fauna for extraction of drugs for medicinal purposes.

The research activities include: systematic collection and identification of marine organisms, extraction and evaluation of medicinal properties of products derived from marine organisms till completion of clinical trials and development of products from the pure molecules.

The nodal Institute, Central Drug Research Institute, Lucknow has been entrusted the responsibility of coordination and implementation; reconfirmation of bioactivity encountered by the other participating institutions, elucidation of chemical structure by way of spectrometry and high throughput screening, testing in laboratory animal models, etc., so as to utilise their international standard infrastructure facilities and their expertise.

The MoES has re-orientated this programme with the focus on exploratory and product development activities and initiated action to induct more institutional participation including pharmaceutical industry.

During the Tenth Plan period (2002-2007), programmes such as "Shoreline Management", "Tidal Inlets Management", "Ecosystem Modelling", "Marine Ecotoxicology", "Application of Ocean Colour Monitoring in sediment transport studies", have been formulated and are being implemented.

Scientists will carry out experiments, particularly climate change. The station will be managed by the NCAOR which also runs India's research stations in Antarctica.

Two-thirds of Ny-Alesund, which is spread over 63,000 sq km, is permanently under ice, but the climate is mild in comparison to other areas so near the North Pole. The

mean temperature in the coldest month of February is minus 14 degrees while in the warmest month of July, it is 5 degrees Centigrade.

Owing to almost zero pollution, Arctic is generally considered by scientists to be better than Antarctica for a wide range of research activities. Antarctica has hosted hundreds of scientists for years now and even attracts some tourists, leading to concerns that problems like pollution would soon catch up with the southernmost continent.

Himadri would initially be manned by Indian scientists on a project- to-project basis and later converted to round-the-year station, like the ones in Antarctica.

Blue Economy:

1. The concept was introduced by Gunter Pauli in his 2010 book- "The Blue Economy: 10 years, 100 innovations, 100 million jobs".
2. It is the sustainable use of ocean resources for economic growth, improved livelihoods and jobs, and ocean ecosystem health.
3. It advocates the greening of ocean development strategies for higher productivity and conservation of ocean's health.
4. It encompasses–
    1) **Renewable Energy:** Sustainable marine energy can play a vital role in social and economic development.
    2) **Fisheries:** Sustainable fisheries can generate more revenue, more fish and help restore fish stocks.
    3) **Maritime Transport**: Over 80% of international goods traded are transported by sea.
    4) **Tourism:** Ocean and coastal tourism can bring jobs and economic growth.
    5) **Climate Change:** Oceans are an important carbon sink (blue carbon) and help mitigate climate change.
    6) **Waste Management:** Better waste management on land can help oceans recover.

5. Blue Economy emphasizes on integration of development of ocean economy with social inclusion, environmental sustainability, combined with innovative business model.

6. This is reflected in **Sustainable Development Goal (SDG 14),** which calls to conserve and sustainably use the oceans, seas and marine resources for sustainable development.

Need for Blue Economy

1. Oceans cover three-quarters of the Earth's surface, contain 97% of the Earth's water, and represent 99% of the living area on the planet.
2. Oceans protect biodiversity, keep the planet cool, and absorb about 30% of global CO_2 emissions.
3. At least 3-5% of global GDP is derived from oceans.
4. Blue economy, through sustainable use of oceans, has great potential for boosting the economic growth by providing opportunities for income generation and jobs etc.
5. It can support food security, and diversification to address new resources for energy, new drugs valuable chemicals, protein food, deep sea minerals, security etc.
6. It is the next **sunrise sector**

Blue Economy for India

1. Blue economy presents India with an unprecedented opportunity to meet its national socio-economic objectives as well as strengthen connectivity with neighbors.
2. Blue Economy can help in focusing on livelihood generation, achieving energy security, building ecological resilience, and improving health and living standards of coastal communities.
3. Blue economy would reinforce and strengthen the efforts of the Indian government as it strives to achieve the SDGs of hunger and poverty eradication along with sustainable use of marine resources by 2030.
4. India has a long coastline of 7,517 km covering nine states and two union territories – with an Exclusive Economic Zone (EEZ) of 2.02 mn. sq.km.

5. Marine services sector could serve as the backbone of its blue economy and help India become 10 trillion-dollar economy by 2022.
6. Indian Ocean is a major conduit of trade with as much as 80% of global oil trade happening through it.
7. Better connectivity in the region will significantly cut the transport cost and maritime wastage of resources making the trade sustainable and cost effective.

Developments Initiated by India

1. The Sagarmala project is the strategic initiative for port-led development through the extensive use of IT enabled services for modernization of ports.
2. India has an umbrella scheme by the name of **O-SMART** which aims at regulated use of oceans, marine resources for sustainable development.
3. **Integrated Coastal Zone Management** focuses on conservation of coastal and marine resources, and improving livelihood opportunities for coastal communities etc.
4. Development of **Coastal Economic Zones (CEZ)** under Sagarmala would become a microcosm of the blue economy, wherein industries and townships that depend on the sea will contribute to global trade.
5. India has a National Fisheries policy for promoting 'Blue Growth Initiative' which focus on sustainable utilization of fisheries wealth from the marine and other aquatic resources.

Questions:

1. Give a detailed answer about Blue Economy with its needs and its initiatives made by government.
2. Explain about the Department of Ocean Development with its mission and vision.
3. Discuss in detail about Polar Explorations by India.